

Choline and betaine intake and the risk of colorectal cancer in men



Dietary choline and betaine have been hypothesized to decrease the risk of cancer because of their role as methyl donors in the one-carbon metabolism. However, it remains unknown whether dietary intake of choline and betaine is associated with colorectal cancer risk. We prospectively examined the associations between dietary choline and betaine intake and risk of colorectal cancer in men in the Health Professionals Follow-up Study. We followed 47,302 men and identified a total of 987 incident colorectal cancer cases from 1986 to 2004. We assessed dietary and supplemental choline and betaine intake every four years using a validated semi-quantitative food frequency questionnaire. The Cox proportional hazards model was used to estimate multivariate relative risks (RRs) and 95% confidence intervals (95% CIs). All statistical tests were two-sided. We did not find any statistically significant associations between choline intake or betaine intake and risk of colorectal cancer. Comparing the top quintile with bottom quintile, multivariate RRs (95% CI) were 0.97 (0.79-1.20; $P_{\text{trend}} = 0.87$) for choline intake and 0.94 (0.77-1.16; $P_{\text{trend}} = 0.79$) for betaine intake. Similarly, we observed no associations between colorectal cancer risk and choline from free choline, glycerophosphocholine, phosphocholine, phosphatidylcholine, or sphingomyelin. Our data do not support that choline and betaine intake is inversely associated with colorectal cancer risk.